

# Lecture Syllabus

## From Plan to Control: Managing a Project Like a Real PMO

AI-Assisted Gamified Project Management Module

### Lecture Information

|                        |                                                                                                                                    |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Module type</b>     | Gamified AI-assisted university lecture                                                                                            |
| <b>Target audience</b> | Third- or fourth-year undergraduate students, or Licence/Master students in engineering, computer science, business, or management |
| <b>Duration</b>        | One 3-hour session, or two 90-minute sessions                                                                                      |
| <b>Main case</b>       | Smart Campus Mobile App project                                                                                                    |
| <b>License</b>         | CC-BY-NC recommended                                                                                                               |

### Lecture Description

In this lecture, students act as a project management office team responsible for delivering a Smart Campus Mobile App in 12 weeks. The project has a limited budget, a small technical team, a required demo by week 6, and a major integration risk with the university database.

Students do not only listen to project management concepts. They define project scope, identify stakeholders, build a Work Breakdown Structure, prepare a schedule, analyze dashboard indicators, face a project crisis, and present a corrective action plan.

### Learning Objectives

By the end of the lecture, students will be able to:

1. Explain the project management cycle: initiation, planning, execution, monitoring, control, and closing.
2. Build a deliverable-oriented Work Breakdown Structure for a realistic project.
3. Convert a WBS into tasks, dependencies, milestones, and deliverables.
4. Estimate duration, cost, resources, and risks.
5. Optimize project execution by adjusting resources, sequencing, and priorities.
6. Track progress using planned value, earned value, actual cost, SV, CV, SPI, and CPI.
7. Diagnose project problems from dashboard data.
8. Propose corrective actions when a project is late, over budget, or technically unstable.
9. Communicate project status in a professional PMO-style briefing.

## Case Scenario

The university wants to build a Smart Campus Mobile App in 12 weeks.

### App Features

- Student login
- Course schedule
- Exam calendar
- Notifications
- Room reservation
- Complaint/request system
- Admin dashboard
- Pilot deployment

### Project Constraints

|                         |                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|
| <b>Budget</b>           | 40,000 USD                                                                                  |
| <b>Duration</b>         | 12 weeks                                                                                    |
| <b>Team</b>             | 1 project manager, 2 backend developers, 2 mobile developers, 1 UI/UX designer, 1 QA tester |
| <b>Main risk</b>        | Integration delay with the university database                                              |
| <b>Sponsor pressure</b> | First demo required by week 6                                                               |

## Gamified Progression

The lecture uses five levels. Each level contains a mission, a deliverable, and a badge.

| Level | Name                | Mission                                                       | Badge                            |
|-------|---------------------|---------------------------------------------------------------|----------------------------------|
| 1     | Project Launch      | Define scope, stakeholders, constraints, and success criteria | Scope Master                     |
| 2     | WBS Builder         | Build a deliverable-oriented WBS                              | WBS Architect                    |
| 3     | Schedule Rescue     | Compress a 15-week plan into a realistic 12-week schedule     | Schedule Optimizer               |
| 4     | Dashboard Detective | Analyze week 6 dashboard data and diagnose project health     | Dashboard Detective              |
| 5     | Crisis Controller   | Respond to the week 8 crisis and present a recovery briefing  | Risk Analyst / Crisis Controller |

## Lecture Schedule

| Time | Activity | Learning Type | Tools |
|------|----------|---------------|-------|
|------|----------|---------------|-------|

|                    |                                       |                            |                                   |
|--------------------|---------------------------------------|----------------------------|-----------------------------------|
| <b>0–15 min</b>    | Opening scenario and decision poll    | Acquisition + discussion   | Genially, Mentimeter/Wooclap      |
| <b>15–45 min</b>   | Planning concepts and project charter | Acquisition + inquiry      | Slides, NotebookLM                |
| <b>45–80 min</b>   | WBS workshop                          | Production + collaboration | Worksheet, Canva, Mermaid/draw.io |
| <b>80–90 min</b>   | Break                                 | Pause                      | –                                 |
| <b>90–120 min</b>  | Schedule optimization challenge       | Practice + collaboration   | Google Sheets/Excel               |
| <b>120–155 min</b> | Dashboard and KPI analysis            | Practice + inquiry         | Google Sheets, Quizizz            |
| <b>155–175 min</b> | Week 8 crisis simulation              | Discussion + production    | Genially, group briefing          |
| <b>175–180 min</b> | Exit ticket                           | Reflection                 | Google Forms or paper             |

## AI Tools Used

| <b>Tool</b>                                 | <b>Role in the Lecture</b>                                                             |
|---------------------------------------------|----------------------------------------------------------------------------------------|
| <b>NotebookLM</b>                           | Source-grounded summaries of project management concepts from uploaded course material |
| <b>ChatGPT / Gemini / Claude</b>            | Scenario writing, prompts, crisis cards, quiz questions, and feedback                  |
| <b>Learning Designer</b>                    | Main module structure, activity types, export, and public link                         |
| <b>Gamma and Canva</b>                      | Slide draft generation and final visual design                                         |
| <b>Napkin AI / Mermaid / draw.io</b>        | WBS, Gantt, RACI, risk matrix, and control-loop diagrams                               |
| <b>Genially</b>                             | Branching simulation and interactive decision screens                                  |
| <b>Quizizz / QuestionWell / MagicSchool</b> | Formative quizzes and instant feedback                                                 |
| <b>Google Sheets / Excel</b>                | Schedule, budget, earned value, and dashboard simulation                               |

## Student Deliverables

Each group submits or presents:

1. Project charter summary
2. Stakeholder list and success criteria
3. Deliverable-oriented WBS
4. Optimized 12-week schedule
5. Risk register or risk notes

6. Dashboard diagnosis
7. Corrective action plan
8. Three-minute PMO recovery briefing

## Assessment

| Criterion                  | Weight | Focus                                                            |
|----------------------------|--------|------------------------------------------------------------------|
| Scope clarity              | 15%    | Stakeholders, constraints, and measurable success criteria       |
| WBS quality                | 20%    | Deliverable orientation, completeness, and logical decomposition |
| Schedule and dependencies  | 20%    | Realistic sequencing, dependencies, and optimization             |
| Risk analysis              | 15%    | Identification of risks and relevant mitigation actions          |
| Tracking and control logic | 20%    | Correct dashboard interpretation and corrective action reasoning |
| Professional presentation  | 10%    | Clear, concise, and convincing PMO briefing                      |

## Preparation

Before the lecture, students should review basic definitions of project scope, stakeholders, WBS, milestone, risk, and dashboard. No advanced project management software experience is required.

## Exit Ticket

At the end of the lecture, students answer:

1. What is one difference between monitoring and controlling?
2. What was the most important corrective action in your group plan?
3. How did AI help your team during the activity?
4. What should a project manager never hide from the sponsor?